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STEP 1: CREATE TABLE.

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| bank     |
| bankdb   |
| db_2520090166 |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
7 rows in set (0.13 sec)

mysql> USE db_2520090166;
Database changed
mysql> CREATE TABLE orders (
  ->   ord_no INT PRIMARY KEY,
  ->   purch_amt DECIMAL(10,2),
  ->   ord_date DATE,
  ->   customer_id INT,
  ->   salesman_id INT
  -> );
Query OK, 0 rows affected (0.24 sec)
```

STEP 2: INSERT DATA

```
mysql> INSERT INTO orders VALUES
-> (70001,150.50,'2012-10-05',3005,5002),
-> (70009,270.65,'2012-09-10',3001,5005),
-> (70002,65.26,'2012-10-05',3002,5001),
-> (70004,110.50,'2012-08-17',3009,5003),
-> (70007,948.50,'2012-09-10',3005,5002),
-> (70005,2400.60,'2012-07-27',3007,5001),
-> (70008,5760.00,'2012-09-10',3002,5001),
-> (70010,1983.43,'2012-10-10',3004,5006),
-> (70003,2480.40,'2012-10-10',3009,5003),
-> (70012,250.45,'2012-06-27',3008,5002),
-> (70011,75.29,'2012-08-17',3003,5007),
-> (70013,3045.60,'2012-04-25',3002,5001);
Query OK, 12 rows affected (0.04 sec)
Records: 12  Duplicates: 0  Warnings: 0

mysql>
```

QUESTION 1 - WHERE CLAUSE

```
mysql> SELECT *
-> FROM orders
-> WHERE purch_amt > 2000;
+-----+-----+-----+-----+-----+
| ord_no | purch_amt | ord_date | customer_id | salesman_id |
+-----+-----+-----+-----+-----+
| 70003 | 2480.40 | 2012-10-10 | 3009 | 5003 |
| 70005 | 2400.60 | 2012-07-27 | 3007 | 5001 |
| 70008 | 5760.00 | 2012-09-10 | 3002 | 5001 |
| 70013 | 3045.60 | 2012-04-25 | 3002 | 5001 |
+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

QUESTION 2 - WHERE CLAUSE

```
mysql> SELECT *
-> FROM orders
-> WHERE ord_date='2012-09-10';
+-----+-----+-----+-----+-----+
| ord_no | purch_amt | ord_date | customer_id | salesman_id |
+-----+-----+-----+-----+-----+
| 70007 | 948.50 | 2012-09-10 | 3005 | 5002 |
| 70008 | 5760.00 | 2012-09-10 | 3002 | 5001 |
| 70009 | 270.65 | 2012-09-10 | 3001 | 5005 |
+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)

mysql>
```

QUESTION 3 WHERE CLAUSE

```
mysql> SELECT *  
-> FROM orders  
-> WHERE salesman_id=5001;
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70002	65.26	2012-10-05	3002	5001
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001

```
4 rows in set (0.00 sec)
```

QUESTION 4 - ORDER BY.

```
mysql> SELECT *  
-> FROM orders  
-> ORDER BY purch_amt DESC;
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70010	1983.43	2012-10-10	3004	5006
70007	948.50	2012-09-10	3005	5002
70009	270.65	2012-09-10	3001	5005
70012	250.45	2012-06-27	3008	5002
70001	150.50	2012-10-05	3005	5002
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70002	65.26	2012-10-05	3002	5001

```
12 rows in set (0.00 sec)
```

```
mysql>
```

QUESTION 5 ORDER BY

```
mysql> SELECT *
      -> FROM orders
      -> ORDER BY ord_date;
+-----+-----+-----+-----+-----+
| ord_no | purch_amt | ord_date | customer_id | salesman_id |
+-----+-----+-----+-----+-----+
| 70013 | 3045.60 | 2012-04-25 | 3002 | 5001 |
| 70012 | 250.45 | 2012-06-27 | 3008 | 5002 |
| 70005 | 2400.60 | 2012-07-27 | 3007 | 5001 |
| 70004 | 110.50 | 2012-08-17 | 3009 | 5003 |
| 70011 | 75.29 | 2012-08-17 | 3003 | 5007 |
| 70007 | 948.50 | 2012-09-10 | 3005 | 5002 |
| 70008 | 5760.00 | 2012-09-10 | 3002 | 5001 |
| 70009 | 270.65 | 2012-09-10 | 3001 | 5005 |
| 70001 | 150.50 | 2012-10-05 | 3005 | 5002 |
| 70002 | 65.26 | 2012-10-05 | 3002 | 5001 |
| 70003 | 2480.40 | 2012-10-10 | 3009 | 5003 |
| 70010 | 1983.43 | 2012-10-10 | 3004 | 5006 |
+-----+-----+-----+-----+-----+
12 rows in set (0.00 sec)
```

QUESTION 6 - SUM()

```
mysql> SELECT SUM(purch_amt) AS total_revenue
      -> FROM orders;
+-----+
| total_revenue |
+-----+
| 17541.18 |
+-----+
1 row in set (0.01 sec)

mysql> _
```

QUESTION 7 - AVG()

```
mysql> SELECT AVG(purch_amt) AS average_order
      -> FROM orders;
+-----+
| average_order |
+-----+
| 1461.765000 |
+-----+
1 row in set (0.00 sec)

mysql>
```

-

QUESTION 8 MAX()

```
mysql> SELECT MAX(purch_amt) AS highest_order
-> FROM orders;
+-----+
| highest_order |
+-----+
|          5760.00 |
+-----+
1 row in set (0.00 sec)

mysql>
```

QUESTION 9 - MIN()

```
mysql> SELECT MIN(purch_amt) AS lowest_order
-> FROM orders;
+-----+
| lowest_order |
+-----+
|          65.26 |
+-----+
1 row in set (0.00 sec)

mysql>
```

QUESTION 10 - COUNT()

```
mysql> SELECT COUNT(*) AS total_orders
-> FROM orders;
+-----+
| total_orders |
+-----+
|           12 |
+-----+
1 row in set (0.35 sec)

mysql>
```

QUESTION 11 GROUP BY

```
mysql> SELECT salesman_id,  
->      SUM(purch_amt) AS total_sales  
-> FROM orders  
-> GROUP BY salesman_id;
```

salesman_id	total_sales
5002	1349.45
5001	11271.46
5003	2590.90
5005	270.65
5006	1983.43
5007	75.29

6 rows in set (0.01 sec)

```
mysql> █
```

QUESTION 12 - GROUP BY

```
mysql> SELECT customer_id,  
->      SUM(purch_amt) AS total_purchase  
-> FROM orders  
-> GROUP BY customer_id;
```

customer_id	total_purchase
3005	1099.00
3002	8870.86
3009	2590.90
3007	2400.60
3001	270.65
3004	1983.43
3003	75.29
3008	250.45

8 rows in set (0.00 sec)

```
mysql> █
```


QUESTION 13 GROUP BY + MAX()

```
mysql> SELECT customer_id,  
->      MAX(purch_amt) AS highest_purchase  
-> FROM orders  
-> GROUP BY customer_id;
```

customer_id	highest_purchase
3005	948.50
3002	5760.00
3009	2480.40
3007	2400.60
3001	270.65
3004	1983.43
3003	75.29
3008	250.45

```
8 rows in set (0.00 sec)
```

QUESTION 14 - GROUP BY + HAVING

```
mysql> SELECT salesman_id,  
->      SUM(purch_amt) AS total_sales  
-> FROM orders  
-> GROUP BY salesman_id  
-> HAVING SUM(purch_amt) > 3000;
```

salesman_id	total_sales
5001	11271.46

```
1 row in set (0.00 sec)
```

```
mysql> _
```

QUESTION 15 GROUP BY + HAVING

```
mysql> SELECT customer_id,  
->      SUM(purch_amt) AS total_purchase  
-> FROM orders  
-> GROUP BY customer_id  
-> HAVING SUM(purch_amt) > 2500;  
  
+-----+-----+  
| customer_id | total_purchase |  
+-----+-----+  
|          3002 |          8870.86 |  
|          3009 |          2590.90 |  
+-----+-----+  
2 rows in set (0.00 sec)  
  
mysql> █
```

QUESTION 16 - GROUP BY + HAVING

```
mysql> SELECT customer_id,  
->      COUNT(*) AS total_orders  
-> FROM orders  
-> GROUP BY customer_id  
-> HAVING COUNT(*) > 1;  
  
+-----+-----+  
| customer_id | total_orders |  
+-----+-----+  
|          3005 |             2 |  
|          3002 |             3 |  
|          3009 |             2 |  
+-----+-----+  
3 rows in set (0.00 sec)  
  
mysql>
```

QUESTION 17 - GROUP BY + HAVING + ORDER BY

```
mysql> SELECT customer_id,  
->      SUM(purch_amt) AS total_purchase  
-> FROM orders  
-> GROUP BY customer_id  
-> HAVING SUM(purch_amt) > 1000  
-> ORDER BY total_purchase DESC;
```

customer_id	total_purchase
3002	8870.86
3009	2590.90
3007	2400.60
3004	1983.43
3005	1099.00

5 rows in set (0.00 sec)

```
mysql>
```

QUESTION 18 - BETWEEN + HAVING

```
mysql> SELECT customer_id,  
->      MAX(purch_amt) AS max_purchase  
-> FROM orders  
-> GROUP BY customer_id  
-> HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
```

customer_id	max_purchase
3002	5760.00
3009	2480.40
3007	2400.60

3 rows in set (0.00 sec)

```
mysql> _
```

QUESTION 19 - COUNT + HAVING

```
mysql> SELECT salesman_id,  
->      COUNT(*) AS total_orders  
-> FROM orders  
-> GROUP BY salesman_id  
-> HAVING COUNT(*) >= 2;
```

```
+-----+-----+  
| salesman_id | total_orders |  
+-----+-----+  
|          5002 |          3 |  
|          5001 |          4 |  
|          5003 |          2 |  
+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql>
```

QUESTION 20 - MAX + GROUP BY + HAVING

```
mysql> SELECT ord_date,  
->      MAX(purch_amt) AS highest_purchase  
-> FROM orders  
-> GROUP BY ord_date  
-> HAVING MAX(purch_amt) > 2000;
```

```
+-----+-----+  
| ord_date   | highest_purchase |  
+-----+-----+  
| 2012-10-10 |          2480.40 |  
| 2012-07-27 |          2400.60 |  
| 2012-09-10 |          5760.00 |  
| 2012-04-25 |          3045.60 |  
+-----+-----+  
4 rows in set (0.00 sec)
```

```
mysql>
```